

DA3408 Assignment 1

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Question 1

1)

- a) **Entanglement:** Changing one part of the code changes everything.
- b) **Undeclared Customers:** The marketing dashboard team.
- c) **Configuration and Glue-Code Debt:** 14 undocumented shell scripts creates tangled pipeline.

2)

For (c), we can track the 14 shell scripts and implement the pipeline using dvc.

Question 2

run_id	metrics.f1_macro	metrics.accuracy	params.hidden_layer_sizes	params.learning_rate_init	tags.mlflow.runName
f7664686ccd647d3a689f53e3e8ebc17	0.972281	0.972500	(100, 50)	0.001	mlp-layers_100x50-lr_0.001
7a66f1b126cb409aae6c982469e15423	0.970960	0.971143	(100,)	0.001	mlp-layers_100-lr_0.001
393113ec74144fc6abcc9e543633bfec	0.970960	0.971143	(100,)	0.001	mlp-baseline
2869b8c88e224673bae2486d7211c09a	0.970785	0.971000	(100,)	0.01	mlp-layers_100-lr_0.01
3a2e48901d264c5785d29c4a70046d92	0.966585	0.966929	(100, 50)	0.01	mlp-layers_100x50-lr_0.01
63881ee7022441a39f075c68c2ae5270	0.965654	0.965857	(50,)	0.01	mlp-layers_50-lr_0.01
347724ea35af4bda98a0fee0c742caa4	0.965156	0.965429	(50,)	0.001	mlp-layers_50-lr_0.001
a9a6e974c80c4eca91e5ae569c135b5e	0.970960	NaN	(100,)	0.001	rf-autolog

Best run: f7664686ccd647d3a689f53e3e8ebc17 (accuracy=0.9725)

We can see that the best accuracy is given by the model with hidden layer sizes (100, 50) and learning rate 0.001. And it is also higher for layer size 100 than 50. So the Accuracy increases with layer size, number of layers and also as we make learning rate smaller, only till an optimal value. From the accuracies, there doesn't seem to be any evidence of overfitting here.

This code is for tracing the parameters:

```
def train_and_log(hidden_layer_sizes=(100,), learning_rate_init=0.001, batch_size=128, max_iter=20, run_name=''):
    with mlflow.start_run(run_name=run_name):
        # --- parameters (at least 3) ---
        mlflow.log_param("hidden_layer_sizes", str(hidden_layer_sizes))
        mlflow.log_param("learning_rate_init", learning_rate_init)
        mlflow.log_param("batch_size", batch_size)
        mlflow.log_param("max_iter", max_iter)

        model, acc, f1 = train_and_evaluate(hidden_layer_sizes, learning_rate_init, batch_size, max_iter)

        # --- metrics (at least 2) ---
        mlflow.log_metric("accuracy", acc)
        mlflow.log_metric("f1_macro", f1)

        mlflow.set_tag("team", "data-science")
        mlflow.sklearn.log_model(model, name="model")

    run_id = mlflow.active_run().info.run_id
    print(f"Logged run {run_id} | acc={acc:.4f} f1={f1:.4f}")
    return run_id
```